Two reaction–diffusion species driving growth and death:

is the mechanism sound as biology, as code, and as Plexus2?

`tyssue_rd_ops` · `tyssue_ops3d` · `make_two_species.py` · `make_sculpt.py` · `log/okuda/{ts,tsd,sc}_*` (24 preliminary runs) · 11 August 2026

Scope. This note is about one addition to the framework: a second reaction–diffusion species, and the two operators that consume it — growth inhibition and cell death. It asks whether that addition is *sound*, on three axes: as biology, as code, and as Plexus2. **It does not ask whether the mechanism achieves the campaign's goal.** Twenty-four preliminary runs exist in `log/okuda` and they raised a specific set of defects; §6 is that list, structured, with each one's status. A mechanism whose implementation is unverified cannot be compared with an observation at all, so the goal question is deferred until this note's gates close.

1. The mechanism

One field cannot pattern two processes. Every run this project has made carries a single Gray–Scott pair, and every operator reads its activator: growth is gated on it, death is gated on it, the purse-string selects on it. So the tissue has exactly one map. "Where it grows" and "where it dies" are then the same place or exact complements of each other, by construction, and no setting of any parameter can separate them. That is a limitation of the *representation*, not of the search.

Two species is two maps. Biologically this is unremarkable and well attested: a tissue carries many morphogens at once, on different length scales, read by different targets. Two are enough to state the smallest interesting case — one field says *grow here*, another says *stop* or *die here* — and that is the case this note verifies.

The two consumers are drawn from different literatures. Growth inhibition is lateral inhibition in the Gierer–Meinhardt sense: a diffusible signal that suppresses the response of the cells around a source. Patterned death is Monier *et al.* (2015), where apoptotic force drives epithelial folding in *Drosophila* — and it is the only *inward* deformation available in this vocabulary, every other operator pushing the sheet outward.

The arithmetic that fixes the representation. A dying cell leaves an epithelium by losing neighbours until it is a triangle; a triangle collapses to a point as a T2 transition, $V-2$, $E-3$, $F-1$, so $\chi = V - E + F$ is unchanged and the sheet stays closed at genus 0. Collapsing a face with $k > 3$ sides would merge k vertices into one and leave a vertex of degree k, **which a trivalent sheet cannot represent**. So death cannot be one operator: it is mark, then shed, then extrude, and the shedding belongs to `reconnect_t1_3d`. That is not a hidden coupling; it is the mechanism. A cell leaves an epithelium by losing neighbours.

Two ratios fix the proportions. The species' wavelengths must differ or the second map is redundant — here by a factor of two in diffusivity, $d_a : 0.08 \to 0.16$. And the clearing time of a dying cell is set by two numbers and nothing else (Eq. (4)).

2. Equations and symbols

Two independent systems in one buffer. `chem` is width 4; species A occupies columns $(0, 1)$ and species B columns $(2, 3)$. Each has its own seeder, diffusion and reaction, and each operator declares which pair it owns through `chan`:

$$\frac{\partial a_X}{\partial t} = D_a^X \nabla^2 a_X + f(a_X, u_X), \qquad \frac{\partial u_X}{\partial t} = D_u^X \nabla^2 u_X + g(a_X, u_X), \qquad X \in \{A, B\}, \tag{1}$$

with f, g Gray–Scott. The two systems are coupled *only* through the operators that read them, never through the buffer — which is a property that must be gated (G7), not assumed.

Growth, and the zero it did not have. The standing law is purely activating,

$$\dot{v}_{\mathrm{eq}} = \texttt{rate} \cdot \big(\rho + \mathrm{hill}(a_A)\big), \qquad \mathrm{hill}(x) = \frac{x^n}{x_{\mathrm{sw}}^n + x^n} \in [0, 1], \tag{2}$$

so the slowest a cell can grow is $\texttt{rate} \cdot \rho$, and $\rho > 0$ is required by premise P1 (a tissue that adds no material is not growing). The inhibitor supplies the missing zero, multiplicatively:

$$\dot{v}_{\mathrm{eq}} = \texttt{rate} \cdot \big(\rho + \mathrm{hill}(a_A)\big) \cdot \big(1 - \mathrm{hill}_{\mathrm{inh}}(\hat{a}_B)\big), \qquad \hat{a}_B = a_B / \max_f a_B. \tag{3}$$

$\hat{a}_B$ is normalised to the inhibitor's own maximum for the same reason a_{sw} is a fraction: an absolute threshold against a field whose scale the chemistry sets is either always on or always off, and this project has paid for that twice.

Death, as three steps and one closed-form time.

$$V_f^0 \leftarrow (1-s)V_f^0, \qquad \text{extrude when } V_f^0 < c\, v_{\text{ref}}, \qquad \boxed{\tau_{\text{clear}} = \frac{\ln c}{\ln(1-s)} \text{ ticks}} \tag{4}$$

Volume is conserved into the survivors, and the conservation is bounded. When f dies its remaining material is shared over its live ring neighbours, and the increment is capped so the recipient stays inside the integrator's basin:

$$c_g \leftarrow c_g + \min\Big(\frac{\text{amt}/|\mathcal{L}|}{V_g^0},\ c_{\max} - c_g\Big), \qquad \mathcal{L} = \{g \in \mathcal{N}(f) : V_g^0 \geq c\, v_{\text{ref}}\}, \tag{5}$$

with $c_{\max}$ the field's own running maximum. Material that will not fit is *dropped and counted* in `apop_spill`. Both restrictions were found by failures and both are gated (G4, G5).

A death RATE, not a death set: with N live cells, $|\{\text{marked}\}| \leq \phi N$, and when the cap binds the worst candidates are taken by the mode's own severity. Geometric modes are exempt, because a set chosen once has no flux.

a_X, u_X	activator, substrate of species X	*per cell*, INTENSIVE. A concentration, so it is meaningful to compare between cells of different size
`chan`	species selector	which `chem` pair an operator owns. Default 0, so every pre-existing spec is unchanged
D_a^X, D_u^X	diffusivities	the RATIO sets the Turing wavelength; the pair sets the speed
ρ	growth baseline	tip:body is $(\rho+1) : \rho$. $\rho = 0$ is an infinite ratio, which P1 refuses
x_{sw}, n	gate threshold, sharpness	a FRACTION of that field's own maximum, never absolute
s, c	`shrink_rate`, `critical_frac`	per tick, and a fraction OF v_{ref}
v_{ref}	reference volume	the SEED-TIME median. Never recomputed: recomputing it after deaths moves every threshold in the model
τ_{clear}	clearing time	INTENSIVE: a property of (s, c) alone, not of the tissue
ϕ	`max_mark_frac`	fraction of LIVE cells under sentence at once — bounds a FLUX
n_{apop}	deaths	EXTENSIVE: scales with the tissue, so never comparable between runs of different size without dividing by N
`apop_spill`	unplaced material	EXTENSIVE. The conservation law's own error term

3. The Plexus2 container

No new set. What is new is a *family* the vesicle lacked — *Die* — and a second species inside one state buffer. Three claims about the container have to hold, and each is a gate:

1. **Two instances of one operator must be ADDITIVE, not overwriting.** `cell_react` returns a delta over the whole `chem` tensor. Written naively it fills columns $(0, 1)$, so a second instance would have overwritten the first rather than driving its own species. Each instance now writes zeros outside its own pair, which is what makes "two species" expressible at all rather than a name for one. (G7)
2. **`apoptosis_3d` is `structural`, not `lateral`.** It changes how many entities exist, so it may not be a pure state map: it renumbers faces, and every per-face array and every *pending delta* must be permuted with it. (G3)
3. **A named population is a seed window, one level up.** `list`, `band` and `cone` choose a population ONCE. Re-evaluating them every frame converts a death sentence into a death rate — the same argument the paper makes for seeding, where an initial condition re-applied every step becomes a moving boundary condition. (G6)

operator	**kind / family**	**reads**	**mechanism**	**gate**
`seed_cell_rd`	seed / growth	—	seeds its own columns	G7
`cell_diffuse`	lateral / fields	`chem`, adjacency	per-column diffusivity vector	G7
`cell_react`	lateral / fields	`chem`	Eq. (1), zeros outside its pair	G7
`grow_3d`	structural / growth	`chem`	Eq. (3)	G8, G9
`apoptosis_3d`	**structural / die**	`chem`, mesh	Eqs. (4)–(5)	G1–G6
`reconnect_t1_3d`	rewire / topology	mesh	sheds neighbours — REQUIRED	G2

Schedule. Death runs *between growth and the mechanics*: growth sets the targets, death overrides them for the sentenced, and the relaxation sees both in one tick, so a cell extruded this frame has its hole closed this frame. Measured, same composition with death appended after the recorder instead: grip 0.031 against 0.110. There is no `substep_dt` block — this vesicle integrates at one rate and `shape_energy_3d` carries `relax_iters` internally.

4. The gates

Thresholds set before the run. The tiers are the paper's: **bookkeeping** asks whether the code does what the operator says, **closed form** whether it reproduces physics it was given, and **measurement** whether it agrees with something observed in cells.

	gate	threshold, before	measured
tier 1 — bookkeeping			
G1	every selector fires on a known population and NOT on a uniform field	all cases	**19/19**
G2	$\chi = 2$ at every frame with death running	exact	**2, all frames**
G3	a topology change permutes pending DELTAS, not only state	activator ≥ 0	**0** (was $-1.04{\times}10^{10}$)
G4	a bequest goes only to neighbours above the extrusion threshold	no divisor $< c\,v_{\mathrm{ref}}$	**P4, P12 cleared**
G5	a bequest leaves the recipient inside the integrator's basin	no NaN, activator ≥ 0	**RE-RUNNING** —
G6	a named population is chosen ONCE; only state modes re-evaluate	one death per named cell	**1, was 7**
G7	two species never write each other's columns	$\{0,1\}$ and $\{2,3\}$	**exact, non-uniforn**
G8	`inhib_chan` absent reproduces a pre-change run	metrics identical	**RUNNING** — §6 I
G9	growth $\to 0$ as the inhibitor saturates	$< 5\%$ of uninhibited	$0.978 \to 0.014$
G10	the cap admits at most ϕN, worst first, and drains	exact	**8/100; took worst;**
G11	n_{apop} equals the cells removed	exact, death-only composition	$400 - N_{\mathrm{final}}$
tier 2 — closed form			
G12	a collapse is a T2: $V-2$, $E-3$, $F-1$	exact, else refuse	**refuses $k > 3$, re-ch**
G13	clearing time equals $\ln c/\ln(1-s)$	within 1 tick	**NOT MEASUREI**
G14	the throughput lever is s, not ϕ	deaths $\propto 1/\tau_{\mathrm{clear}}$	$\phi{\times}50{\to}{\times}1.7$; $s{\times}3{\to}$
G15	a geometric chisel removes exactly its declared population	exact	**5/5 geometries**
G16	the second species' wavelength differs from the first	spot count differs	**NOT MEASUREI**
tier 3 — measurement			
G17	apoptotic rate matches an epithelium's	1–3% of cells per day	BLOCKED: no `unit`
G18	clearance time matches an extrusion in vivo	30–90 min	BLOCKED: no `unit`
G19	the two species' wavelengths bracket a real morphogen pair	—	BLOCKED: no `unit`

Eleven bookkeeping, five closed form, three measurement — and all three of the last are blocked for want of a `units:` block, the same defect the `ecm` study found when "0.82 grid cells" turned out to be 15 μm. So this note can establish that the operators do what they say and reproduce the arithmetic they were given. **It cannot yet say anything about cells.** Declaring units is the single change that would move three gates from blocked to failable.

5. What the preliminary runs were

Twenty-four runs in three series, all built on `r001_03`: five two-species variants including two controls, six on a death ladder, seven spatial-death "chisels", five inhibitor strengths and five slow-death variants. They were exploratory and are not evidence for the mechanism; they are the source of §6.

Two of their results are worth keeping because they bear on soundness rather than on morphology. **Where death happens dominates how much**: `sc_crown` removed 113 cells and moved `reduced_volume` to 0.814, while `tsd_max` removed 3460 and reached only 0.957 against a 0.959 control — scattered death is erosion, patterned death is a cut, and that is a statement about the operator. **And the inhibitor is monotone**: 12692 → 6920 → 4859 → 4154 cells as `inhib_sw` tightens, which is the behaviour Eq. (3) asserts and is therefore a check on it.

6. Issues the preliminary runs raised, and their status

	issue	what it is, and what was done
I1	`ts_death_b_late` and `ts_death_b_sharp` came back BIT-IDENTICAL	a_{sw} 0.25 vs 0.5, 301 deaths and every metr rate cap binds before the threshold does, could not matter. A rail, found by two rur the last digit. Documented; the cap is now be the throughput lever (G14).
I2	`tsd_cap10`, `tsd_cap25` drove the activator to $-7.3{\times}10^{11}$ and $-2.4{\times}10^{23}$, then NaN	Not caused by the amount of death: `tsd_ca` FIVE TIMES the deaths and is clean. The t the SLOW ones. Conservation is correct as and unbounded as concentration, and Gray–S stably only inside its own basin. Eq. (5) increment and counts the spill.
I3	the inhibitor's null was never run	"off by default changes nothing" was an ass on reading the guard. A pre-change spec is comparison.
I4	the slow-death series is confounded	every variant carries `inhib_sw` 0.35, whic the tissue 12692 → 4859 cells, so those five inhibition and not death speed. My const The series must be re-run with the inhibitor
I5	the two species' wavelengths were never compared	B was GIVEN twice A's diffusivity to be co measured whether it is. `n_spots` per specie and neither is recorded per species.
I6	`sc_old_core` broke P1 and crushed the tissue to 1585 cells	killing the oldest cells removes the body faste replaces it. Behaving as specified; the premi it. Not a defect — a mode that is only usab
I7	an empty `death` dict silently added `apoptosis_3d` at its default mode	four variants meant to isolate INHIBITIO have carried an unasked-for death rule and would have been attributable. Fixed before
I8	`apop_spill` is new and unmeasured	the conservation law's own error term now run has reported it. It should be near zerc run and is the diagnostic for I2.

7. What this licenses, and what it does not

Licensed. Two RD species coexist in one buffer without leaking (G7), which is the structural claim the whole addition rests on. The Die family is sound as bookkeeping: selectors fire where they should and nowhere else (G1), the sheet stays closed (G2), a topology change carries its pending deltas (G3), the count is exact (G11), the rate is a rate (G10), and a named population is a seed window rather than a per-frame rule (G6). Growth can be switched off (G9), and the collapse is a genuine T2 (G12).

Not licensed, and the list is not short. Three gates are open and two are re-running. **G5** — the conservation bound — is the one that matters most, because until it closes the operator can drive the chemistry out of its basin and a run's numbers are then about an integrator. **G13** and **G16** mean two quantities this note reasons with, the clearing time and the second species' wavelength, are asserted rather than measured. **No gate is a measurement** (G17–G19 blocked), so nothing here has been compared with a cell.

The mechanism should not enter the loop until G5, G8 and G13 close. G5 because a mechanism that can NaN the chemistry will produce runs the campaign records as evidence; G8 because an operator that changes runs which do not use it corrupts every comparison in the ledger; G13 because the clearing time is the quantity every death experiment is denominated in. G16 and I4/I5 are about the experiments, not the operators, and can close afterwards.

8. References

References

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§1.
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